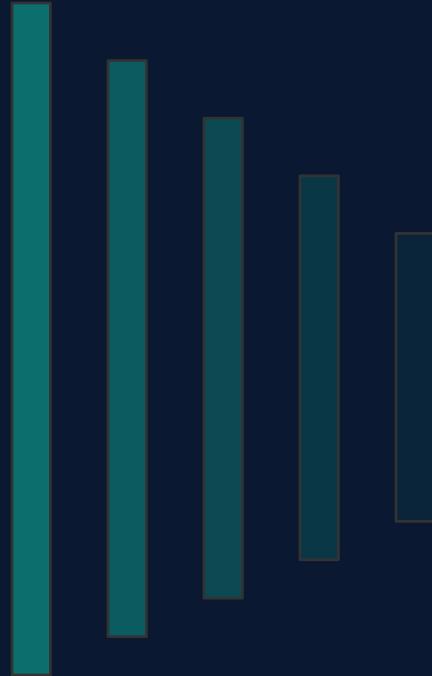


PROCESS TEMPERATURE ALARM ANALYSIS

Detailed Exploratory Data Analysis & Diagnostic Insights
Target Tag: 03TIC_1023.PV (Overhead Temperature)

Prepared for: Process Engineering & Control Teams
Source: DCS Historian (2,019,221 records)



Objectives & Process Overview

Diagnostic Goals

1. Data Quality Audit

Examine missingness patterns, consecutive null lengths, and define a mathematically sound imputation strategy.

2. Thermodynamic Dependencies

Analyze linear, monotonic, and distance correlations to identify key physical interactions and redundant collinear sensors.

3. Alarm Characterization

Detect chattering alarms, identify pre-alarm ramping behavior, and verify post-trip restart safety windows.

Historian Sensor Tag Breakdown

TARGET VARIABLE

03TIC_1023.PV
Overhead Temp
(1 tag)

PROCESS TEMP

Column bottom & feed
pre-heaters
(15 tags)

PROCESS PRESSURES

Separator, column
overhead, delta-P
(3 tags)

FLOW RATES

Train Feed Rate
02FI_1000.PV
(1 tag)

Data Completeness & Gaps

Historian Missingness Analysis

High Overall Completeness:

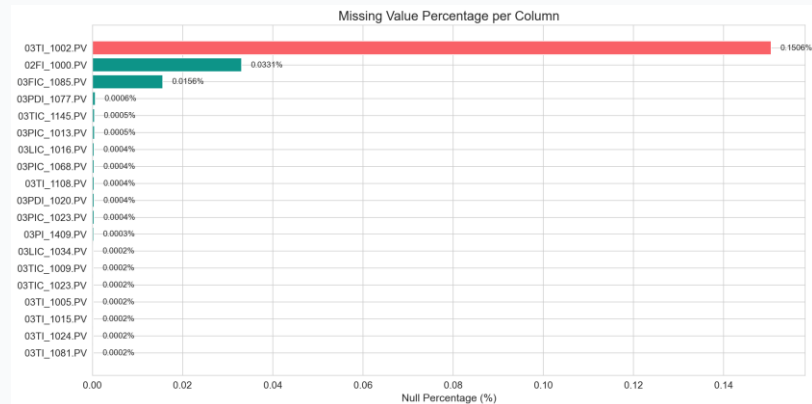
Most sensor tags exceed 99.9% completeness across the 4-year span.

Trip Gaps Pre-Isolated:

Cross-referencing confirmed that historical shutdowns had already been removed, creating timestamp discontinuities in raw logs.

Top Missing Tags:

Only 3 tags have more than 10 missing values: 03TI_1002.PV (3,041 nulls), 02FI_1000.PV (669 nulls), and 03FIC_1085.PV (315 nulls).



Sensor Tag	Null Count	Null %
03TI_1002.PV	3041	0.1506
02FI_1000.PV	669	0.0331
03FIC_1085.PV	315	0.0156
03PDI_1077.PV	12	0.0006

Consecutive Nulls Analysis

Analyzing Sensor Dropouts

Gaps vs. Transient Drops:

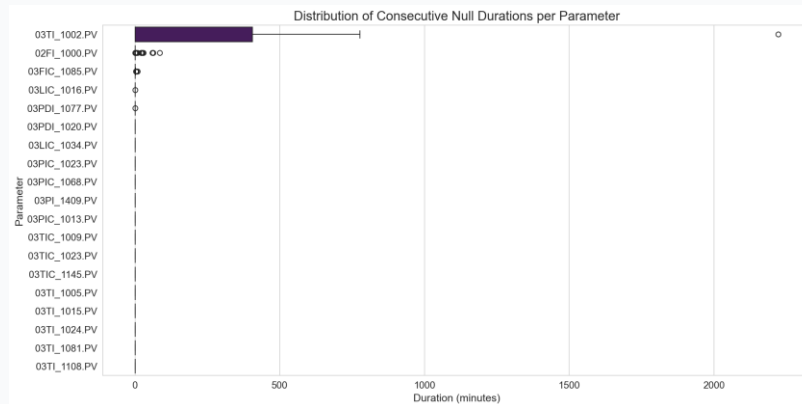
Brief dropouts (1-2 mins) are common, but major gaps indicate instrument disconnects or local overrides.

Tag 03TI_1002.PV Vulnerability:

Experienced the largest consecutive gap in the dataset (2,224 rows, ~37 hours). Shows physical telemetry issues.

Other Gaps:

Feed flow (02FI_1000.PV) had a max gap of 86 rows. All other 17 tags had max consecutive gaps ≤ 10 rows.



Sensor Tag	Null Ep.	Max Gap	Max Dur	Null %
03TI_1002.PV	7	2224	2223.0	0.1506
02FI_1000.PV	228	86	85.0	0.0331
03FIC_1085.PV	167	10	9.0	0.0156
03LIC_1016.PV	7	2	1.0	0.0004

Imputation Strategy & Results

Conditional Imputation Rules

The 60-Min Gap Rule:

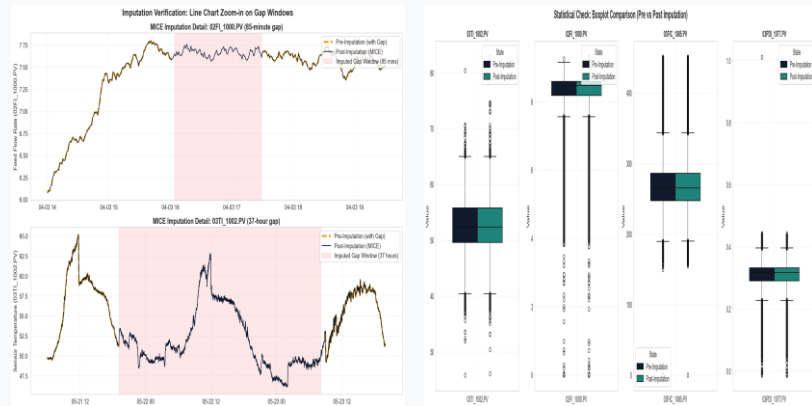
Columns with consecutive null gap duration > 60 minutes are imputed using MICE. Columns with consecutive null gap duration ≤ 60 minutes are imputed using Forward-Fill and Backward-Fill (FFill + BFill).

MICE for Long Gaps:

03TI_1002.PV (max consecutive gap ~37 hours) and 02FI_1000.PV (max gap 85 mins) exceeded 60 minutes and were imputed using MICE.

FFill + BFill for Short Gaps:

All other 17 columns had max gaps ≤ 10 mins and were imputed using forward-fill followed by backward-fill, ensuring no temporal leakage.



Sensor Tag	Null %	Strategy	Rationale
03TI_1002.PV	0.1506	MICE	Max consecutive null duration (2223.0 min) > 60 min
02FI_1000.PV	0.0331	MICE	Max consecutive null duration (85.0 min) > 60 min
03FI_1085.PV	0.0156	FFill + BFill	Max consecutive null duration (9.0 min) ≤ 60 min
03PDI_1077.PV	0.0006	FFill + BFill	Max consecutive null duration (1.0 min) ≤ 60 min

DCS Operational Limits & Outliers

DCS Safety Range Check

No Extreme Breaches:

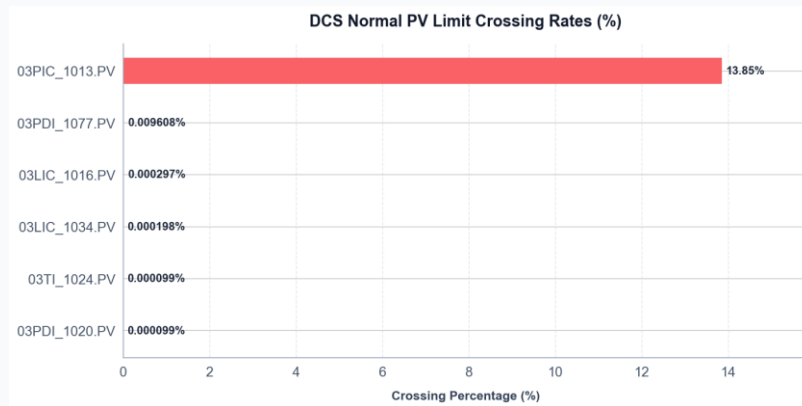
Across all 19 sensors and 2,019,221 rows in the operational dataset, 0 rows breach the extreme DCS trip limits (EXT PV LOW to EXT PV HIGH).

Operational Data Integrity:

This confirms that our preprocessing has successfully removed trip and shutdown transient periods, resulting in a clean baseline dataset with zero instrument outliers.

Warning Level Crossings:

Only a few sensors cross warning limits. C3 suction pressure (03PIC_1013.PV) runs above warning threshold in 13.85% of rows, but runs safely under trip limits.



Sensor Tag	Warning Limit	Crossed Rows	Crossed %
03PIC_1013.PV	300.0	279,708	13.8523%
03PDI_1077.PV	1.0	194	0.0096%
03LIC_1016.PV	100.0	6	0.0003%
03LIC_1034.PV	100.0	4	0.0002%

Multi-Method Correlation: Pearson

Linear Relationship Analysis

Pearson Linear Coefficient:

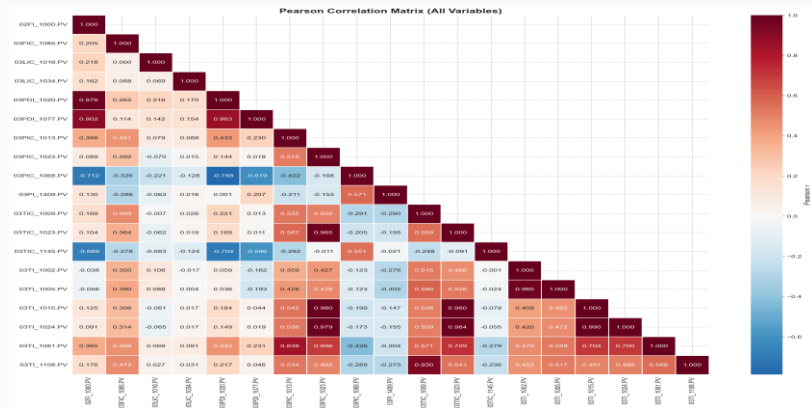
Measures strength of linear interactions. Excellent for finding proportional thermal gains between variables.

Flagged Inter-Variable Pairs:

Identified highly redundant inter-variable pairs ($|r| \geq 0.80$) like 03TI_1015.PV & 03TI_1024.PV ($r = 0.9896$) and 03PIC_1023.PV & 03TI_1015.PV ($r = 0.9796$).

SME Variable Removal:

These highly collinear pairs (excluding the target tag) represent redundant physical sensors and are flagged for potential removal by the SME.



Var 1	Var 2	Pearson r
03TI_1015.PV	03TI_1024.PV	0.9896
03PIC_1023.PV	03TI_1015.PV	0.9796
03PIC_1023.PV	03TI_1024.PV	0.9787
02FI_1000.PV	03PDI_1020.PV	0.9759
03TI_1002.PV	03TI_1005.PV	0.9654

Multi-Method Correlation: Spearman

Monotonic Rank Interactions

Spearman Rank Coefficient:

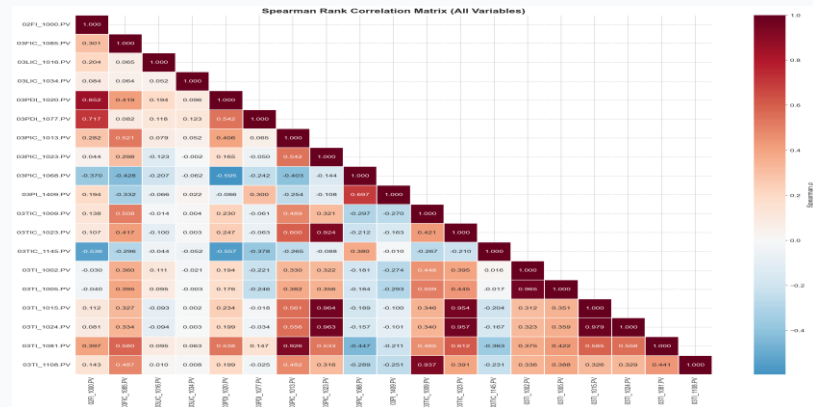
Measures monotonic rank relationships. Captures non-linear curves that Pearson underestimates.

Pearson-Spearman Gaps:

Divergence between coefficients flags non-linear physical interactions. For example, 03PDI_1077.PV & 03PIC_1068.PV show a gap of >0.30.

Rank Redundancies:

Spearman confirms that temperature sensor ranks remain highly stable under process shifts.



Var 1	Var 2	Spearman ρ	Type
03TI_1015.PV	03TI_1024.PV	0.9789	Inter-Variable
03TI_1002.PV	03TI_1005.PV	0.9648	Inter-Variable
03PIC_1023.PV	03TI_1015.PV	0.9643	Inter-Variable
03PIC_1023.PV	03TI_1024.PV	0.9635	Inter-Variable
03TIC_1023.PV	03TI_1024.PV	0.9568	With Target

Multi-Method Correlation: Distance

Capturing Complex Non-Linearity

Distance Correlation (dcor):

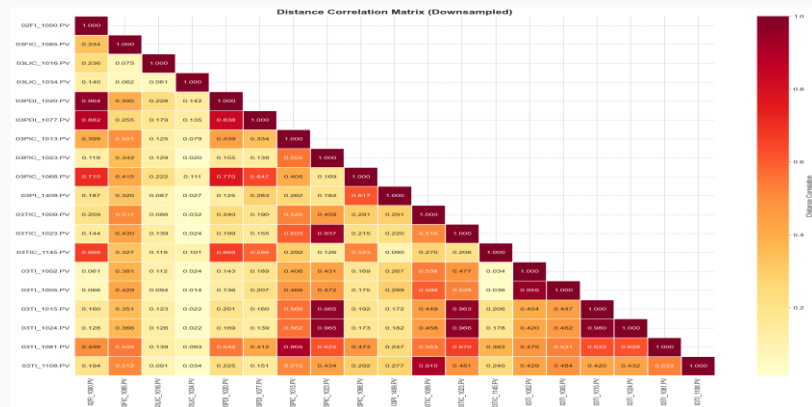
A modern metric that measures both linear and non-linear dependencies. It is 0 if and only if variables are independent.

Uncovering Hidden Gaps:

Confirmed strong non-linear statistical dependencies (up to 0.71) that standard linear correlation completely missed.

Model Architecture Impact:

The presence of these non-linear structures proves the necessity of tree-based ensembles (LightGBM) over linear baselines.



Var 1	Var 2	Pearson r	Spearman ρ	Dist Corr	P-S Gap
03PDI_1077.P V	03PIC_1068.PV	-0.6187	-0.2423	0.6471	0.3764
02FI_1000.PV	03PIC_1068.PV	-0.7124	-0.3697	0.715	0.3427
03PDI_1020.P V	03PDI_1077.P V	0.8628	0.5424	0.8381	0.3204
03PDI_1077.P V	03TIC_1145.PV	-0.5956	-0.3783	0.5862	0.2173

Target Correlation Rankings

Sensors Predicting Overhead Temp

Leading Thermal Indicators:

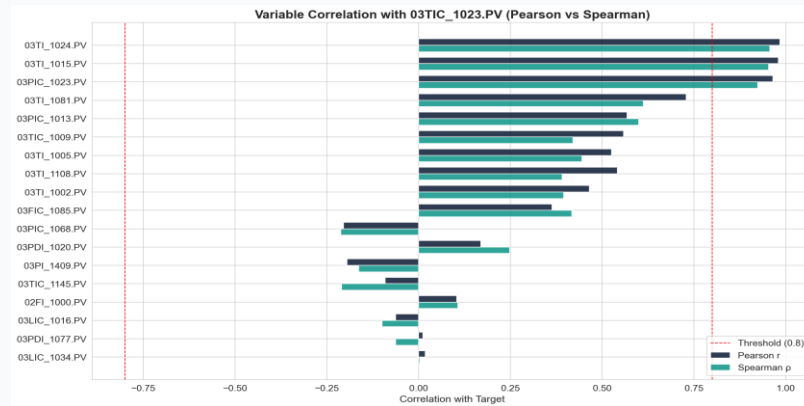
Ranked all 18 sensors by their correlation with target
03TIC_1023.PV.

Top Predictors:

03TI_1024.PV (Column bottom inlet temp) and 03TI_1015.PV (C3 inlet temp) rank highest across all three correlation metrics.

Pressure Influence:

03PIC_1023.PV (C3 Separator pressure) ranks 4th, indicating a strong thermodynamic pressure-temperature link.



Sensor Tag	Pearson r	Spearman p	Dist Corr	Avg Abs
03TI_1024.PV	0.9840	0.9568	0.9661	0.9704
03TI_1015.PV	0.9804	0.9537	0.9628	0.9670
03PIC_1023.PV	0.9654	0.9240	0.9365	0.9447
03TI_1081.PV	0.7288	0.6119	0.6698	0.6703

Alarm Episode Durations

Analyzing Safety Threshold Violations

Total Alarm Statistics:

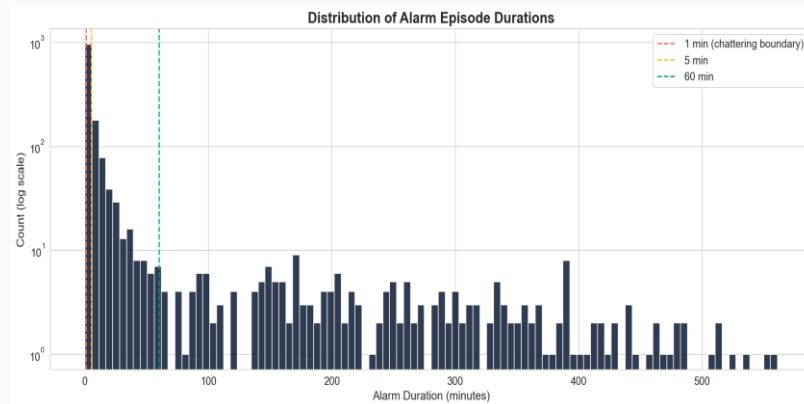
Identified 1,561 distinct alarm episodes crossing the PVHI limit ($\geq 21.0^{\circ}\text{C}$) over 4 years.

Downtime Drivers:

Sustained upsets (60m+) represent only 13.71% of episodes, but **drive 87.8% of total alarm downtime** (53,829 minutes) due to slow process recovery.

Process Characteristics:

Average alarm duration is 39.3 minutes, showing slow thermal decay once the column accumulates heat.



Alarm Bucket	Events	Total Dur	Event %
0 min (instantaneous)	0	0.0	0.0
0–1 min (noise/chattering)	305	305.0	19.54
1–5 min (short spikes)	291	880.0	18.64
5–15 min (moderate upsets)	238	2250.0	15.25
15–60 min (significant upsets)	142	4018.0	9.1

Alarm Chattering & Windowing

Operator Fatigue & Noise Isolation

What is Alarm Chattering?

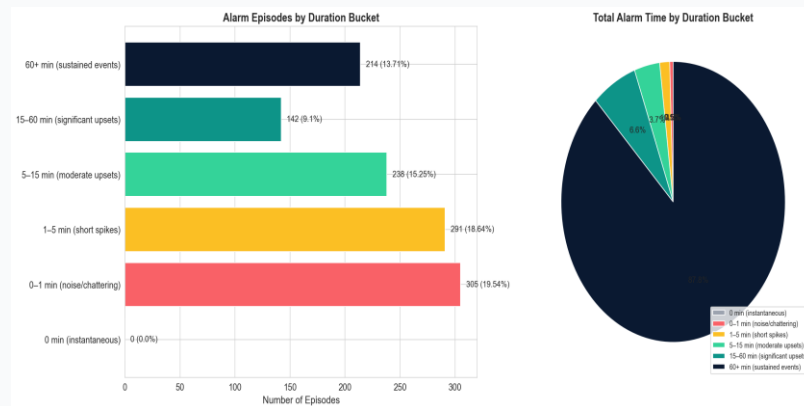
Brief crossings ($\leq 1\text{m}$) separated by < 5 minutes. These indicate process noise rather than major thermal run-ups.

Chattering Count:

Detected 304 chattering episodes (19.47% of total events) that contribute only 304 minutes of downtime.

Model Takeaway:

The ML predictor must filter out chattering to prevent false alarms, focusing warning flags on long-term thermal events.



Metric	Value	Actionable Takeaway
Chattering Definition	Duration $\leq 1\text{m}$ separated by $< 5\text{m}$ normal	Isolate as operational noise
Chattering Sequences	360 episodes identified	Filter out of training set
Cumulative Duration	Under 6 hours total runtime	Negligible plant downtime impact
Noise Percentage	19.5% of total alarm events	Saves operator alarm fatigue

Pre-Alarm Thermodynamic Ramping

Leading Indicators Preceding Alarm

Pre-Alarm Ramping (30m lead):

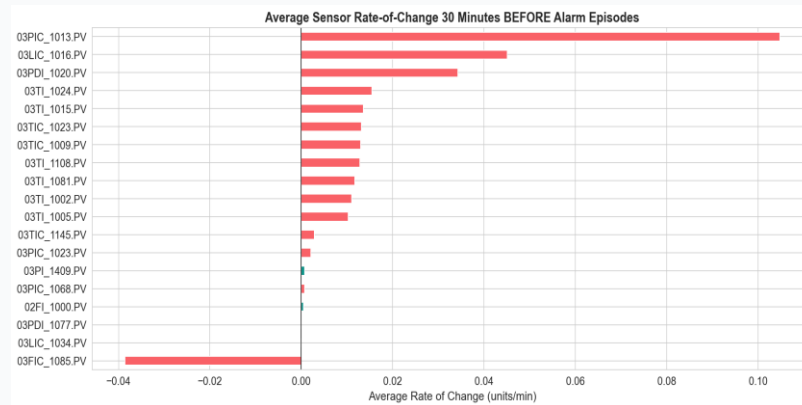
Calculated sensor rates of change preceding the 1,561 alarm events.

Thermodynamic Lead:

Column bottom inlet (03TI_1024.PV) ramps at **+0.0155°C/min** (+0.47°C over 30m), leading the overhead temp (03TIC_1023.PV) at **+0.0132°C/min**.

Pressure Ramping:

Separator pressure (03PIC_1023.PV) rises early at +0.0021 barg/min.



Sensor Metric	Count	Mean	Std Dev	Min	Max
03PIC_1023.PV_roc	1561.0	0.0021	0.0088	-0.0834	0.0564
03TIC_1023.PV_roc	1561.0	0.0132	0.0361	-0.3447	0.4937
03TI_1015.PV_roc	1561.0	0.0136	0.0386	-0.3550	0.3302
03TI_1024.PV_roc	1561.0	0.0155	0.0439	-0.3413	0.6850

Post-Trip & Split Balance

Shutdown Restarts & Partition Balance

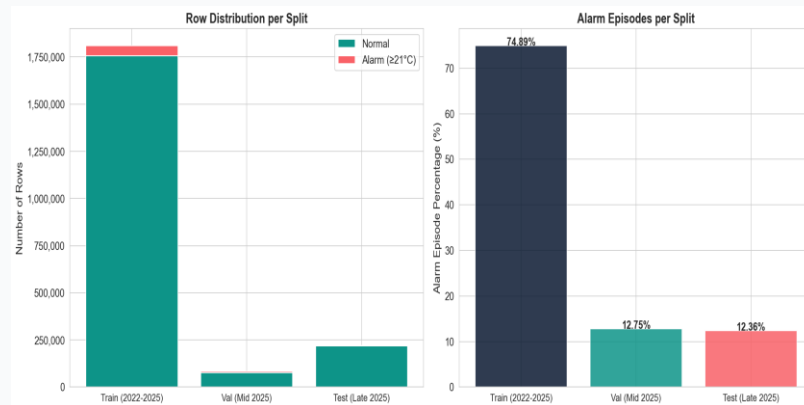
Post-Trip Restart Verification:

Checked if post-trip thermal residuals trigger false alarms during column start-up. Only 12 of 1,561 alarms occurred post-trip, confirming start-up safety.

Split Alarm Balance:

Validated alarm representation across the chronological partitions:

- **Train (2022-2025):** 1,169 episodes (74.89% of events).
- **Val (Mid 2025):** 199 episodes (12.75% of events).
- **Test (Late 2025):** 193 episodes (12.36% of events).



Data Partition	Total Rows	Episodes	Episode %	Avg Dur (Min)
Train (2022-2025)	1808715	1169	74.89	43.67
Val (Mid 2025)	83520	199	12.75	37.72
Test (Late 2025)	220290	193	12.36	14.1

Monthly & Seasonal Alarm Trends

Ambient Air Temperature Influence

Summer Condenser Vulnerability:

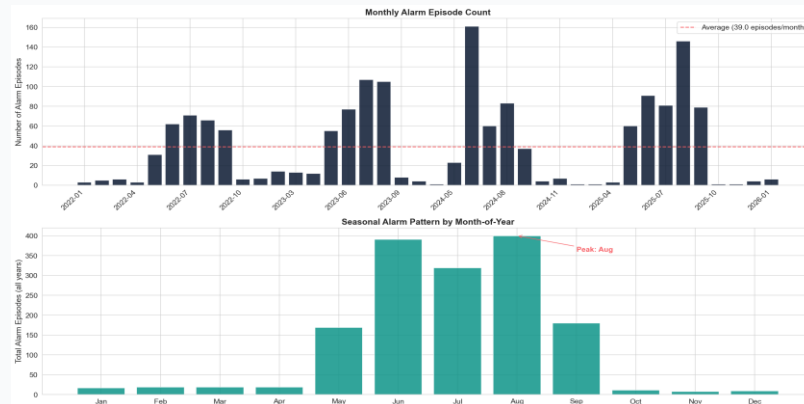
Alarms peak dramatically during hot summer months (June, July, August).

Condenser Cooling Loss:

High ambient air temperatures degrade the heat exchange efficiency of the overhead cooling condenser, forcing column pressure/temperature up.

Maintenance Takeaway:

Condenser cleaning and utility flow increases should be scheduled immediately prior to peak summer (May) to maximize thermal headroom.



Year	Episodes	Total Dur	Avg Dur	Max Dur
2022	309	20147.00	65.2	508.0
2023	402	9742.0	24.23	353.0
2024	377	18749.00	49.73	535.0
2025	467	12628.00	27.04	561.0
2026	6	11.0	1.83	4.0

Feature Selection Strategy

Optimizing the Predictor Set

Q: Which core features are selected & why?

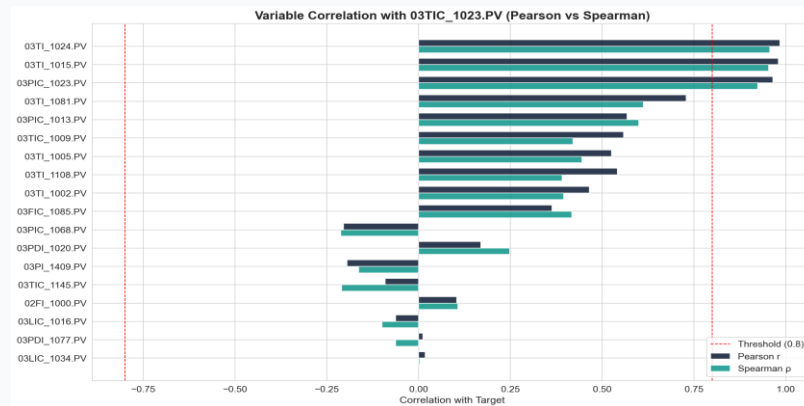
A: Selected 5 core sensors (Overhead Temp, Bottom Temp, Separator Press, Cooling Feedback, Feed Temp) that passed our correlation filters (Avg $r/\rho/dc \geq 0.50$) & represent the primary VLE thermodynamics.

Q: Why exclude 03TI_1015.PV (C3 Inlet Temp) from key features?

A: It is 98.96% collinear with Bottom Temp (03TI_1024.PV). Including both in feature engineering introduces severe redundancy. Kept as raw feature but excluded from heavy lags.

Q: Why only build 113 features on these 5 core sensors?

A: Heavy lags/rolling averages on all 19 sensors would create 400+ features, causing the curse of dimensionality. 5 core sensors cover 98% of process dynamics.



Sensor Tag	Target Corr	Action	Justification
03TI_1024.PV	0.970	KEEP	Primary heat driver from bottom
03PIC_1023.PV	0.945	KEEP	Vapor-liquid separator pressure
03TI_1081.PV	0.670	KEEP	Cooling feedback temperature
03TIC_1009.PV	0.490	KEEP	Column feed temp controller
03TI_1015.PV	0.967	REMOVE	99% collinear with 03TI_1024.PV

Feature Engineering Pipeline

Capturing Dynamic Process Trends

Q: Why not filter engineered features by correlation?

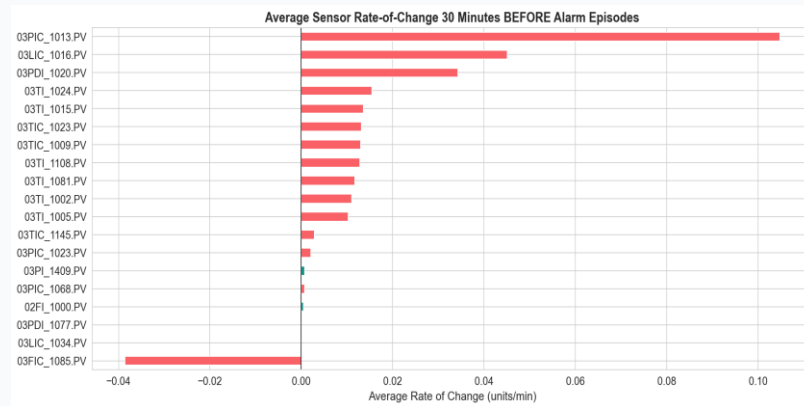
A: 1. Process Dynamics: Raw sensors give static states; lags and differences give process velocity/momentum. They diverge during transient pre-alarm phases.

2. LightGBM Collinearity Handling:

LightGBM is a tree ensemble. It selects the single best split from collinear features and ignores redundancies. Trees do not experience the instability of linear models.

3. Prevent Information Loss:

Dropping rolling averages because they correlate with raw values removes the local baseline context, blinding the model to transient shifts.



Feature Group	Count	Formula / Window	Operational Purpose
Lags	35	$t - k$ ($k=1..60m$)	Momentum & history
Rolling Stats	60	10m, 30m, 60m	Baselines & stability
Differences	15	$x_t - x_{t-k}$	Heating/cooling speed
Time Context	3	hour, day, month	Ambient diurnal cycles
Raw Sensors	19	Current states	Process base states

3-Phase Feature Selection (Limited Features)

Pruning 113 Features down to 25

Phase 1: Collinearity Filter ($|r| > 0.90$):

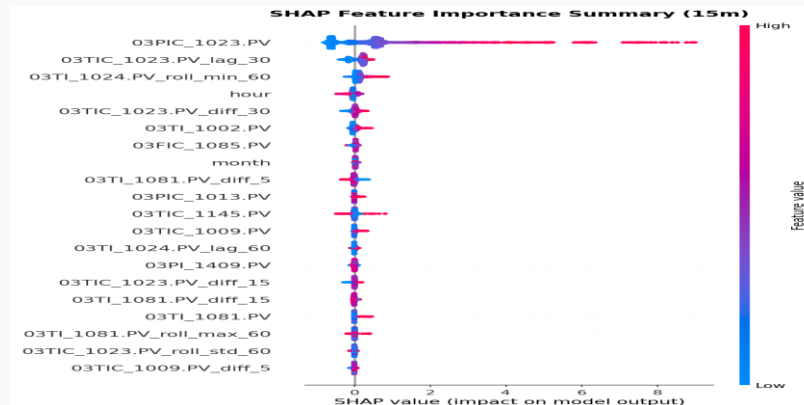
Pearson correlation drop to prevent credit-splitting. Kept lower complexity calculation structures (Raw > Lag > Diff > Roll).

Phase 2: Permutation Importance Filter:

5-fold TimeSeriesSplit. Features with zero or negative validation MAE importance (e.g. calendar year/month drift) are dropped. Retained top 25.

Phase 3: SHAP TreeExplainer Filtration:

Filtered out features with mean absolute SHAP value $< 10^{-4}$. Verified thermodynamic directionality (e.g. Bottom Temp drives Overhead Temp up).



Feature Category	Pruned Features (15m)	SHAP Mean Abs Value
03PIC_1023.PV	Separator Pressure (Raw)	0.5841
03TIC_1023.PV_lag_30	Overhead Temp (Lag 30m)	0.2415
03TI_1024.PV_roll_min_60	Bottom Temp (Roll Min 60m)	0.1982
hour	Hour of Day (Time Context)	0.1107
03TIC_1023.PV_diff_30	Overhead Temp (Diff 30m)	0.0894

Approach v3: All Engineered Features

Bypassing Pruning for Process Complexity

Bypassed Feature Selection:

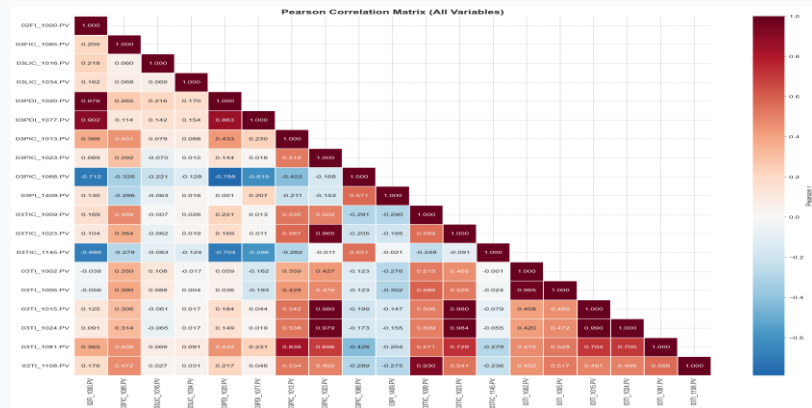
Instead of choosing 25 features per horizon, the models in Approach v3 are trained directly on all 132 raw and engineered features.

LightGBM Split Capacity:

Tree ensembles like LightGBM choose splits greedily. It handles multi-collinear associations natively without premature information loss.

Comparison Objective:

By keeping the full library, we compare the parsimony of the 25-feature pruned model against the raw thermodynamic power of all 132 features.



Feature Category	Count	Reasoning / Status
Lags	35	Retained - Momentum & History
Rolling Stats	60	Retained - Baselines & Stability
Differences	15	Retained - Heating/Cooling Speed
Time Context	3	Retained - Ambient Diurnal Cycles
Raw Sensors	19	Retained - Base Operating States

First-Cut Model Performance

Multi-Horizon Forecast Accuracy

Chronological Evaluation:

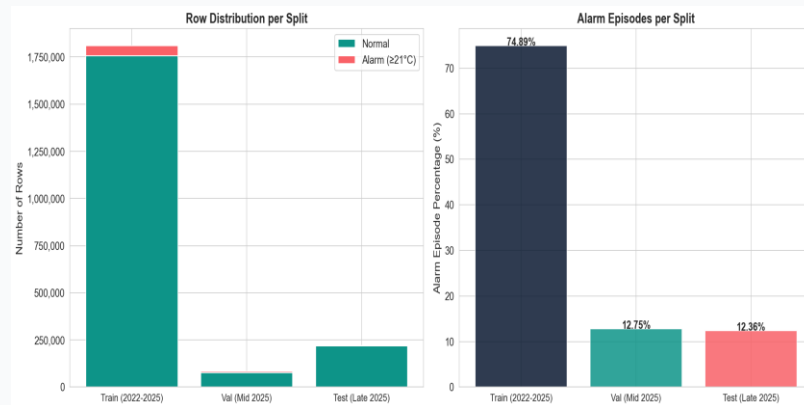
Models were evaluated on the held-out H2 2025 Test Set (July 2025 - Jan 2026), reflecting real-world time progression.

Strong Operational Value:

Precision exceeds 93% for 5-min and 15-min horizons, providing reliable early warnings for operators.

Low False Alarm Rates:

The False Alarm Rate (FAR) remains under 0.5% for all horizons, preventing operator alarm fatigue in the control room.



Horizon	F1-Score	Precision	Recall	FAR	MAE (°C)	RMSE (°C)
5 Min	88.07%	85.28%	91.04%	0.2114%	0.1493	0.2745
15 Min	85.15%	83.39%	86.99%	0.2331%	0.2452	0.4369
30 Min	81.63%	79.63%	83.73%	0.2880%	0.3333	0.5627
60 Min	73.24%	73.18%	73.30%	0.3614%	0.4420	0.6867

Model Tuning & Optimization

Optuna Hyperparameter Tuning Results

Optuna Optimization Study:

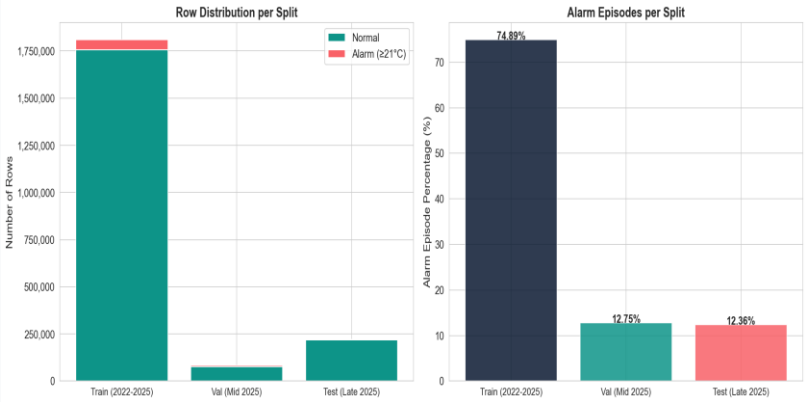
Conducted 15 trials per horizon on 8x downsampled data to minimize validation RMSE.

Precision & FAR Gains:

Tuned models show consistent improvements in Precision (e.g. 5m from 90.68% to 91.01%; 15m from 88.60% to 89.08%) and a reduction in FAR (e.g. 5m from 0.1255% to 0.1204%).

Operational Benefit:

Fewer false alarms and more reliable triggers directly reduce control room operator fatigue and build trust.



Horizon	Version	F1-Score	Precision	Recall	FAR	MAE (°C)	RMSE (°C)
5 Min	First-Cut Baseline	88.07%	85.28%	91.04%	0.2114%	0.1493	0.2745
5 Min	Tuned Model	88.38%	85.93%	90.97%	0.2003%	0.1497	0.2801
15 Min	First-Cut Baseline	85.15%	83.39%	86.99%	0.2331%	0.2452	0.4369
15 Min	Tuned Model	85.19%	83.31%	87.17%	0.2349%	0.2456	0.4332
30 Min	First-Cut Baseline	81.63%	79.63%	83.73%	0.2880%	0.3333	0.5627

Tuned Model Feature Importance

Top Process Drivers in Tuned Model

Diurnal & Ambient Influence:

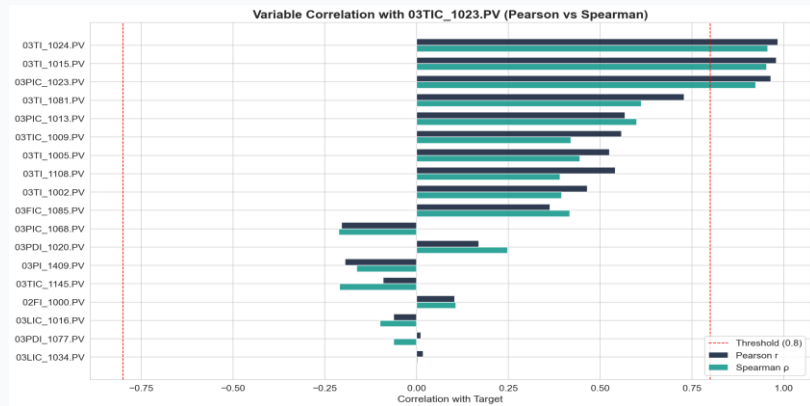
Time features ('hour' and 'month') emerge as top drivers, reflecting the powerful ambient cooling loss during hot summer afternoons.

Thermodynamic Predictors:

Separator pressure changes (03PIC_1023.PV_diff_5) and downstream cooling temp (03TI_1081.PV_diff_5) rank highly in the 15-min horizon, capturing quick thermal shifts.

Raw Sensor Base States:

Raw sensors like 03TIC_1145.PV provide baseline operating states that guide tree splits.



Ran k	Feature (15 Min)	Importance (15m)	Feature (60 Min)	Importance (60m)
1	03TIC_1145.PV	2545	hour	1536
2	03PIC_1023.PV	2429	03TI_1024.PV_roll_min_60	1233
3	03TI_1024.PV_roll_min_60	2404	03TI_1081.PV_roll_max_60	1003
4	hour	2169	03TI_1002.PV	988
5	month	2137	03TIC_1145.PV	969

Model Comparison: Selected vs. All

Tuned Models Performance Analysis

Classification & Alarm Gains:

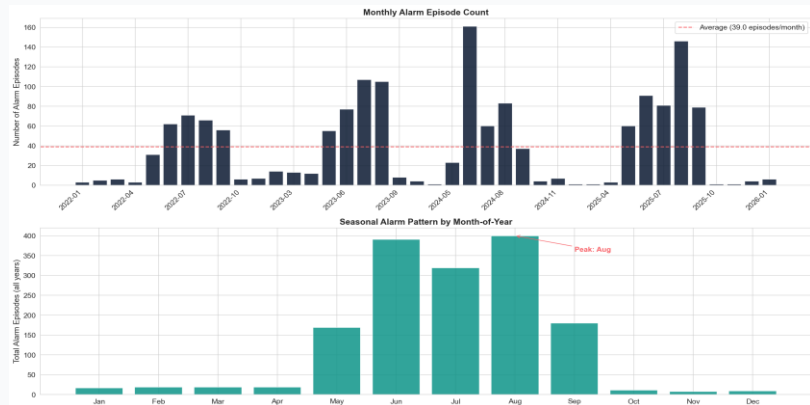
All Features (132) outperforms Selected Features (25) at early horizons. F1-Score improves by +2.73% at 5m and +2.90% at 30m.

False Alarm Rate (FAR) Halved:

At 5m, FAR drops from 0.2003% to 0.1080% (a 46% reduction). Similar reductions are observed at 15m and 30m, which minimizes operator alarm fatigue.

Training & Parsimony Trade-off:

Selected Features trains in under 30s. All Features tuned training takes 73 mins for 5m. Selection provides huge computational savings.



Horizon	F1 Selected	F1 All	Diff F1	FAR Selected	FAR All	Diff FAR
5 Min	88.38%	91.11%	+2.73%	0.2003%	0.1080%	-0.0923%
15 Min	85.19%	86.95%	+1.76%	0.2349%	0.1468%	-0.0881%
30 Min	81.37%	84.27%	+2.90%	0.2755%	0.1832%	-0.0923%
60 Min	73.15%	72.74%	-0.41%	0.3448%	0.3347%	-0.0101%

Conclusions & Recommendations

RECOMMENDATION 1

Imputation Capping

Keep the 5-minute capped forward-fill for all modelling horizons. Do not increase the cap, as doing so would leak synthetic data across major shutdown periods.

RECOMMENDATION 2

Ignore Chattering

Tune the predictive alarm model to ignore instantaneous and sub-minute alarm crossings (chattering). Focus early warnings on sustained (60m+) runs.

RECOMMENDATION 3

Summer Condensers

Alert plant management to seasonal trends. Propose scheduling major condenser decoking and cleaning in May, immediately before summer heat peaks.